

Directional and Tunable End-Effector and Root Compliance for Humanoid Loco-Manipulation

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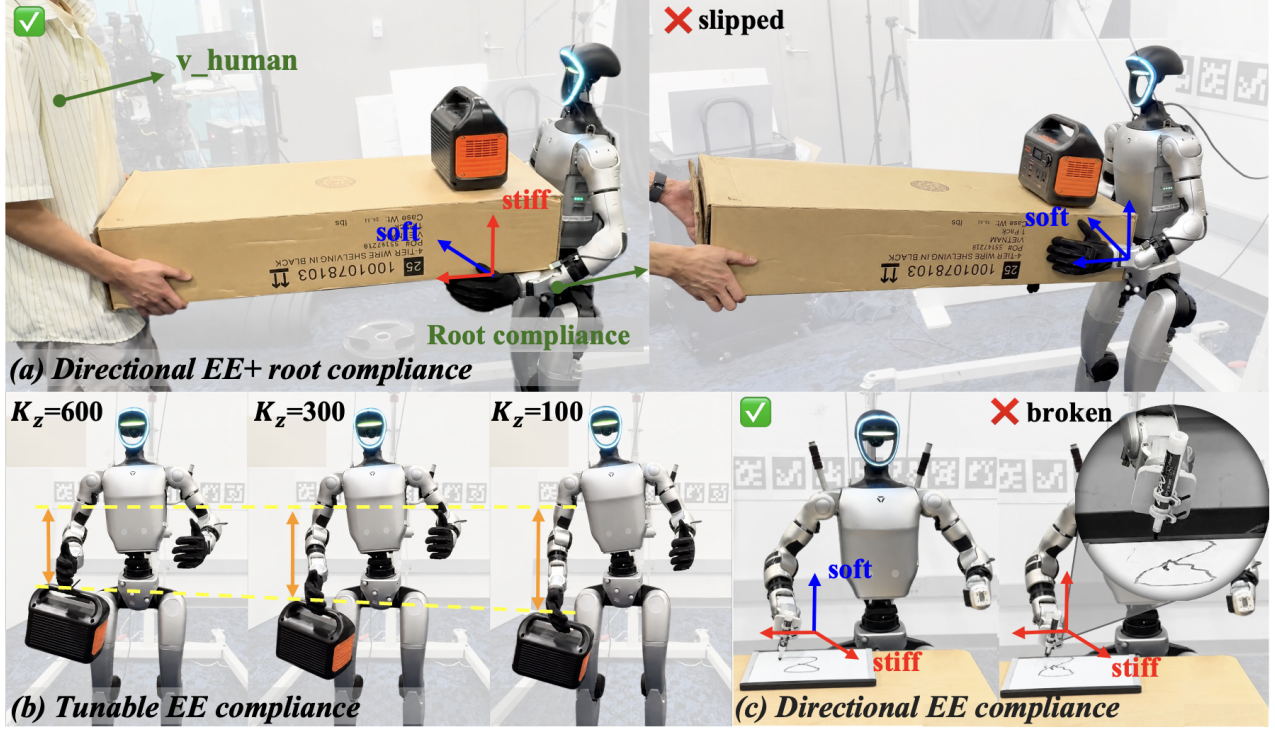


Fig. 1: Our framework enables directional and tunable EE compliance together with three root modes: resist, damping $B=60$, and $B=200$ damping. (a) Directional EE and root compliance support collaborative box carrying, while an isotropically soft setting causes the payload to slip. (b) Online changes in K_z produce different vertical EE displacements while dragging a 3 kg payload. (c) Directional EE compliance enables accurate yet compliant writing, whereas excessive stiffness can damage the contact tool.

Abstract—Humanoids are increasingly capable of tracking complex whole-body motions, but physical interaction introduces a different challenge. When a robot makes contact with a person or the environment, it needs to respond to external forces while preserving the motion needed for the task. This response can vary across directions in the end-effectors and on the body. For example, an end effector may need to accommodate contact force in one direction while maintaining motion accuracy in another, while the robot body may resist an external force or move with it. We present a compliance framework for humanoid loco-manipulation that combines directional and tunable end-effector (EE) compliance with selectable root compliance for external force rejection or force following. A hierarchical reinforcement learning controller modulates a fixed whole-body tracking policy through high-level EE and root commands, while interaction forces are estimated from proprioceptive history. Our simulation and real-world experiments on a humanoid demonstrate directional stiffness control, online stiffness adjustment, distinct root compliance, compliant manipulation, and collaborative carrying. Our project page is at: <https://ee-root-compliance.github.io/>

I. INTRODUCTION

Recent advances in humanoid whole-body control have established increasingly capable policies that can serve as general behavioral foundations. Feature-based dense tracking methods [1], [2] support scalable motion tracking and provide interfaces for high-level planning, whereas generative-adversarial based models [3] improve recovery and transitions across diverse motions. These capabilities allow humanoids to execute complex whole-body motions with increasing accuracy. However, during physical interaction, continuing to track the reference motion without responding to external forces can generate excessive interaction forces or destabilize the robot. Accurate motion tracking alone does not specify how a humanoid should respond when physically interacting with humans or the environment.

Recent compliant and force-aware humanoid controllers [4]–[10] improve physical interaction beyond motion tracking, but they do not jointly specify directional, tunable EE compliance and root response. Simply making an EE isotropically soft can reduce contact force, but it sacrifices

accuracy in task-relevant directions. Writing, for example, requires low stiffness normal to the surface and high in-plane stiffness, while different contacts may also require different stiffness magnitudes. In addition, different loco-manipulation tasks have different specifications on whether the body should resist, yield, or follow the external force. More generally, how to achieve whole-body compliance for a floating-base humanoid remains an open problem.

In this work, we take a step toward structured humanoid compliance by organizing it around two primary functional components: **manipulation** and **locomotion**. For manipulation, we introduce directional EE compliance that is continuously tunable over 100–600 N/m, allowing different Cartesian stiffness values to be commanded along different axes. For locomotion, we represent compliance at the root through one resistance mode and two distinct damping modes. Together, they provide a structured interface in which EE stiffness and root response can be adjusted independently and applied concurrently during physical interaction.

Realizing these behaviors with a single policy is challenging because training across external forces and axis-wise stiffness commands creates substantial multi-task interference, while deployment without wrist force/torque sensors requires force estimation. We therefore use a two-stage hierarchical RL pipeline: a stiff low-level whole-body tracking policy is trained first and then held fixed while high-level residual policies and a force estimator are jointly trained through teacher–student learning with privileged force observations. At deployment, an analytical compliance-space mixture-of-experts (MoE) composes stiffness-specific EE experts, while a root-mode command selects one of three root policies. This design separates compliance learning from motion tracking and reduces interference among tasks with different stiffness requirements. Simulation evaluations show that the hierarchy and analytical MoE improve full-range directional EE compliance and realize distinct root responses, while real-world tasks demonstrate online stiffness adjustment and concurrent EE–root compliance. In summary, our contributions are:

- We introduce a structured formulation of humanoid compliance spanning both manipulation and locomotion: directional and continuously tunable EE compliance for manipulation, and one resistance mode and two damping modes at the root for locomotion.
- We develop a two-level hierarchical RL controller in which high-level command residuals realize softer EE responses and root compliance on top of a fixed, stiff low-level whole-body tracking policy. An analytical compliance-space MoE composes 10 EE experts to provide independently adjustable stiffness along each Cartesian axis over 100–600 N/m.
- We validate the proposed compliance definition through real-world manipulation and loco-manipulation tasks, demonstrating the utility of directional and tunable EE compliance together with root compliance in physical interaction.

TABLE I: Comparison of humanoid compliance methods.

Method	Direction	Angular	Tunable	Gen. Track.	Scope	Loco.
Gentle [5]	×	×	100–300 [†]	✓	Upper body	×
CHIP [7]	×	×	1/ k : 0–0.05 [‡]	✓	EE	×
SoftMimic [4]	×	×	40–1000	×	EE	×
CEER [6]	×	×	×	✓	EE	×
LAC [11]	×	✓	10–500	✓	Arms + torso	×
Ours	✓	×	100–600	✓	EE + root	✓

All unmarked tunable stiffness ranges are in N/m. [†]Gentle controls a force limit of 5–15 N; the reported 100–300 N/m corresponds to the reference-dynamics stiffness derived from $K_p = \tau_{\text{safe}}/0.05$, rather than measured Cartesian stiffness. [‡]CHIP controls the compliance coefficient $1/k$ in the range 0–0.05 m/N, rather than directly commanding Cartesian stiffness; $1/k = 0$ corresponds to the stiff limit.

II. RELATED WORK

A. Compliance Control for Humanoid Robots

Recent works have introduced compliant behaviors into humanoid control from different perspectives. GentleHumanoid [5] focuses on force-limited compliant tracking, CHIP [7] introduces tunable end-effector compliance, and SoftMimic [4] uses IK-based compliant motion augmentation to generate feasible whole-body reference responses, enabling balanced and posture-preserving compliance. CEER [6] provides a reusable end-effector compliance interface built on a general whole-body tracking policy. More recently, LAC [11] extends compliance to multiple upper-body links and introduces explicit angular compliance, while Minimalist Compliance Control [10] combines estimated interaction forces with analytical task-space admittance control. Table I summarizes these methods along several dimensions. “Direction” denotes independently controllable directional Cartesian stiffness, “Gen. Track.” denotes general motion-tracking capability, “Scope” indicates where compliance is explicitly defined, and “Loco.” denotes an explicit root compliance formulation for locomotion. Our method is the only one to explicitly support directional compliance. It also provides a wide tunable stiffness range while retaining general tracking capability. More importantly, our formulation covers both end-effector manipulation and root-level locomotion responses, providing a broader and more functionally structured formulation of humanoid compliance.

B. Force- and Tactile-Aware Humanoid Interaction

Beyond explicit compliance control, recent works incorporate interaction forces or tactile feedback into policy learning. Force-adaptive methods use physical interaction information to improve whole-body robustness and coordination under external forces [12]–[16]. For manipulation, tactile sensing and force-aware control have also been integrated into learned humanoid policies [8], [17], [18]. Other works focus on estimating external contacts or wrenches from proprioception and robot dynamics, providing interaction information that can subsequently support force-aware control [9]. These approaches make force and contact information available for interaction control; however, sensing or estimating interaction does not itself specify distinct compliance behaviors.

C. Contact-Aware Humanoid Loco-Manipulation

Complementary to methods that reason about interaction forces, contact-aware loco-manipulation methods explicitly represent contact relationships between the robot, objects, and the environment. One line of work focuses on preserving contact structure during motion retargeting or imitation. OmniRetarget [19], HDMI [20], and InterMimic [21] transfer human-object interaction patterns to humanoids while maintaining physically meaningful contact relationships. Another line introduces contact information directly into the policy interface. SceneBot [22], OmniContact [23], and ContactMimic [24] condition whole-body control on explicit contact labels, schedules, or body-part contact commands, enabling runtime control over where and when contacts occur. Geometry-aware approaches such as LessMimic [25] further encode interaction through distance fields and local geometric features rather than predefined contact labels.

Together, these lines of work advance humanoid compliance control, contact force estimation, and motion generation with contact constraints. Our work complements these methods by specifying *how* the robot responds to contact through directional and tunable EE compliance and selectable root compliance.

III. METHOD

We use a two-level hierarchy with a low-level whole-body tracking policy and high-level compliance policies to reduce multi-task interference across compliance targets during learning. The key idea is to use a sufficiently stiff low-level whole-body tracking policy as the fixed backbone, allowing high-level compliance policies to realize a range of softer behaviors through command residuals. We train separate high-level compliance policies for different compliance settings. We further find that the EE compliance experts can be analytically composed in compliance space to provide continuous, independently adjustable stiffness along each Cartesian axis, while the root compliance policies are selected through mode switching.

We first define directional and tunable EE compliance. We define locomotion compliance in terms of root resistance and damping modes. Then, we describe the two-stage training procedure and the deployment-time composition and switching of the high-level compliance policies (Fig. 2).

A. Directional and Tunable EE Compliance

We define directional EE compliance for each EE $h \in \{L, R\}$ through the desired Cartesian displacement under an external force. This force is expressed in the robot root frame as $\mathbf{f}_{h,t}^b = (f_{h,x,t}^b, f_{h,y,t}^b, f_{h,z,t}^b)$. Given the nominal EE position command $\mathbf{x}_{h,t}^{\text{nom},b}$ and commanded Cartesian stiffness $\mathbf{K} = \text{diag}(K_x, K_y, K_z)$, the target displacement is $\Delta \mathbf{x}_{h,t}^{*,b} = (f_{h,x,t}^b/K_x, f_{h,y,t}^b/K_y, f_{h,z,t}^b/K_z)$ and $\mathbf{x}_{h,t}^{*,b} = \mathbf{x}_{h,t}^{\text{nom},b} + \Delta \mathbf{x}_{h,t}^{*,b}$.

B. Root Resistance and Damping Modes

We define locomotion compliance through the root frame's motion in response to external forces. We implement three

root compliance modes: a resistance mode that rejects external forces, a low-damping mode that performs force following with damping of $B = 60 \text{ Ns/m}$, and a high-damping mode of force following with $B = 200 \text{ Ns/m}$. For the damping mode, given the nominal root velocity command $\mathbf{v}_{xy,t}^{\text{nom},w}$, damping coefficient B , and net force $\mathbf{f}_{xy,t}^{w,\text{net}}$, the target world-frame velocity is $\mathbf{v}_{xy,t}^{*,w} = \mathbf{v}_{xy,t}^{\text{nom},w} + \mathbf{f}_{xy,t}^{w,\text{net}}/B$. These relations define the compliance targets used to train the high-level residual policies. Note that we use damping rather than stiffness to define the force following behavior: the robot should keep moving while being pushed or pulled, rather than stopping at an offset position relative to a fixed equilibrium. This formulation also aligns with the low-level policy's planar velocity command interface.

C. Stage 1: Low-Level Whole-Body Tracking Policy Training

The low-level policy is a stiff whole-body tracking controller that serves as the backbone for high-level compliance control. It receives robot-state observations and an EE-root target and outputs joint actions that coordinate the whole-body to track these targets while maintaining stability.

We train the low-level policy with a DeepMimic-style dense motion-tracking objective [26] and a three-phase teacher-student pipeline [5]. We adopt the EE-root command interface and its corresponding tracking objectives [6]. During training, external disturbance forces are applied at upper-body links while the original motion references remain unchanged, encouraging the policy to track stiffly under perturbations. We curate AMASS [27] using the LIMMT data screening criteria [28] and train a general whole-body tracking policy conditioned on EE and root commands. Our default three-keypoint (3-kp) interface commands the root and two EEs; the five-keypoint (5-kp) interface commands the root, two EEs, and two feet.

D. Stage 2: High-Level Residual Policy Training

To achieve desired compliance control behaviors, the high-level compliance policies output EE or/and root command residuals, which are added to the nominal EE-root command to form the control targets provided as input to the low-level whole-body tracking policy.

1) *Residual Actions and Training Objectives*: In Stage 2, the low-level policy is frozen, and each high-level residual policy is trained with PPO. The EE residual action consists of six Cartesian position offsets, with one xyz offset for each EE. Each normalized action is passed through tanh and scaled by an expert-specific per-axis limit before it is added to the nominal EE command. The 100 N/m soft-axis experts use a 0.35 m limit to cover the required displacement range; the other experts use 0.15 m. Root compliance policies output root-command residuals; feet and yaw residuals are optional action-space extensions. Let $\mathbf{x}_{h,t}^b$, $\mathbf{v}_{xy,t}^w$, and $\mathbf{p}_{xy,t}^w$ denote the actual EE position, root planar velocity, and root planar position, respectively, with $\mathbf{p}_{xy,0}^w$ denoting the initial root position. Superscripts b and w indicate the robot root and world frames, and xy indicates planar components. The

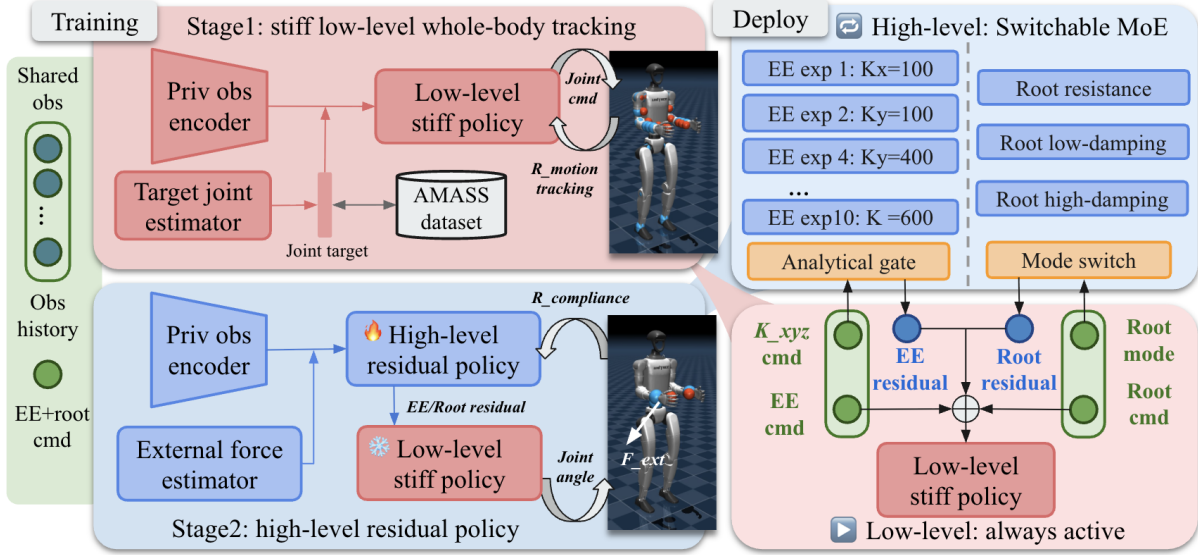


Fig. 2: System overview. Stage 1 trains a stiff low-level whole-body tracking policy with an EE-root command interface. In Stage 2, the low-level policy is held fixed while high-level residual policies learn EE or root command corrections that produce the desired response to external forces under commanded stiffness or a selected root mode. At deployment, an analytical compliance-space gate composes 10 EE compliance experts according to the commanded Cartesian stiffness, while a root compliance policy is selected through mode switching. The low-level policy remains active throughout, enabling online adjustment of EE stiffness and root mode switching.

task errors and rewards are

$$\begin{aligned}
 e_t^{\text{EE}} &= \frac{1}{2} \sum_{h \in \{L, R\}} \|\mathbf{x}_{h,t}^b - \mathbf{x}_{h,t}^{*,b}\|_2, \\
 e_t^{\text{D}} &= \|\mathbf{v}_{xy,t}^w - \mathbf{v}_{xy,t}^{*,w}\|_2, \\
 e_t^{\text{R}} &= \|\mathbf{p}_{xy,t}^w - \mathbf{p}_{xy,0}^w\|_2, \\
 r_t^m &= \exp(-e_t^m / \sigma_m), \quad m \in \{\text{EE}, \text{D}, \text{R}\}.
 \end{aligned} \tag{1}$$

Where D and R denote damping and resistance, $\sigma_m > 0$ is the error scale for objective m , and the factor $1/2$ averages over the two EEs. The full objective also penalizes large and temporally non-smooth command residuals.

2) *EE Compliance Experts*: An EE compliance expert receives the nominal EE command, recent commands and actions, and robot-state history. It changes only the EE position; the nominal EE orientation remains in the low-level command. We train 10 experts: one isotropic 600 N/m expert and directional 100, 200, and 400 N/m experts for each Cartesian axis. We train a separate stiffness-specific expert because our experiments showed that a single policy trained across multiple stiffness tends to produce similar force-displacement responses despite different commanded stiffness.

3) *Root Compliance Policies*: We train three root compliance policies: one resistance policy that minimizes planar root displacement and two damping policies that track force-dependent velocities with $B = 60$ and $B = 200$ Ns/m. Each is trained as a separate high-level residual policy.

4) *Teacher-Student Training*: In Stage 2, teacher-student training proceeds in two phases. First, a teacher actor is trained with privileged force and simulation-state observations, while a privileged-latent estimator learns from deployable observation histories. The teacher actor is then copied to the student and kept fixed while this estimator continues

to match the teacher latent. In addition to the standard privileged-latent estimator, we train a force-estimation head with direct force supervision. For the EE policies, this head predicts the two root-frame hand forces; for the root policies, it predicts the net force and a body-location label. At deployment, the student uses only its observation history and the estimated latent and force features.

E. Deployment-Time Policy Composition

At deployment, an analytical compliance-space MoE composes the EE compliance experts according to the commanded Cartesian stiffness, while a root-mode command selects the resistance, low-damping, or high-damping policy. This deployment scheme enables online EE stiffness adjustment, root-mode switching, and combined EE-root compliance through the fixed low-level policy.

1) *Analytical Compliance-Space MoE*: At deployment, the analytical MoE composes the EE experts trained in Stage 2. For a commanded stiffness K_d on axis d , we define compliance as $C_d = 1/K_d$. Since the ideal displacement satisfies $\Delta x_d = f_d/K_d = f_d C_d$, the outputs of the two neighboring anchor experts are interpolated linearly in compliance space. The composed output is $\Delta x_{h,d}^{\text{MoE}} = \sum_{k \in \mathcal{N}_d} w_{d,k} \Delta x_{h,d}^{(d,k)}$, where $\mathcal{N}_d$ contains the neighboring stiffness anchors and the weights $w_{d,k}$ sum to one. Each Cartesian axis is composed independently, enabling separate commands for K_x , K_y , and K_z . The 100–600 N/m range is set by the apparent stiffness of the low-level policy and the achievable residual displacement (Sec. IV-B). Because the interpolation weights are determined directly by $C_d = 1/K_d$, the gate has no learned parameters.

2) *Root Policy Selection*: At deployment, a root-mode command selects one of the three policies trained in Stage 2. Unlike the EE experts, the root policies are not interpolated

because the root response depends on foot-ground contacts and support conditions.

3) *Combined EE-Root Control*: The analytical MoE adds an EE command residual and the selected root policy adds a root command residual to the nominal EE-root command. Because the two residuals modify different command components, EE and root compliance can run simultaneously without a priority rule or policy reset.

IV. EXPERIMENTS

Our experiments address three main questions: (i) how accurately the proposed framework realizes commanded directional EE stiffness and root compliance modes; (ii) whether the two-level hierarchy and analytical compliance-space MoE improves accurate directional control over the full stiffness range; and (iii) whether and why the combination of directional and tunable EE compliance with root compliance improve real-world loco-manipulation tasks.

We first describe the perturbation protocol, low-level policy, and force and stiffness ranges. We then evaluate the EE hierarchy, full-range compliance, and workspace consistency, followed by root modes and action spaces. Finally, we demonstrate combined EE-root compliance in real-world loco-manipulation.

A. Experimental Setup and Evaluation Protocol

All policies are trained in Isaac Sim with 16,384 parallel environments on eight L40S GPUs. The low-level (LL) policy uses 7 billion environment steps and takes approximately 6 hours. Each high-level (HL) policy uses 5 billion environment steps without fine-tuning and takes approximately 2 hours. The stiffness-conditioned HL baseline is trained for 10 billion environment steps to ensure convergence. Real-world experiments use a Unitree G1, with policy inference running on an RTX 4080 laptop connected to the robot via Ethernet. No wrist force/torque sensors are used.

We evaluate EE compliance in simulation. In each trial, each EE is commanded to a fixed initial pose. External forces are independently applied to the corresponding left and right hands along the positive and negative x -, y -, and z -directions, with magnitudes of $\{5, 10, 15, 20, 30\}$ N. This results in $2 \text{ hands} \times 3 \text{ axes} \times 2 \text{ directions} \times 5 \text{ force magnitudes} = 60$ trials for each evaluated stiffness configuration. Each force is ramped from zero over 0.5 s and then held for 2.0 s, giving 2.5 s of force exposure. Measurements are averaged over the final 0.5 s of the hold period.

EE compliance-tracking error is the Euclidean distance between the actual EE position and the compliant target. **Nominal EE tracking error** is the Euclidean distance between the actual EE position and the nominal EE command. **Apparent stiffness** is estimated from the ratio of the applied force to the resulting EE displacement along each Cartesian axis. Additional range metrics are introduced in Sec. IV-C.2.

All learned policies are evaluated as deployable students without privileged inputs. Unless otherwise specified, all hierarchical methods use the same stiff 3-kp low-level policy.

TABLE II: Low-level tracking-policy and training-force characterization. Apparent stiffness values describe natural behavior and are not errors relative to a target.

Policy	Nom. Err. (m) ↓	Median (N/m)	P95 (N/m)
3-kp stiff (30 N)	0.0456 ± 0.0140	621.4	1793.0
CEER [6]	0.1385 ± 0.1657	103.3	285.0
SONIC [1]	0.0955 ± 0.0482	192.3	342.4
3-kp stiff (40 N)	0.0524 ± 0.0148	544.7	961.8
3-kp stiff (50 N)	0.0528 ± 0.0134	672.1	3426.0
3-kp stiff (60 N)	0.0830 ± 0.0269	406.5	1088.8

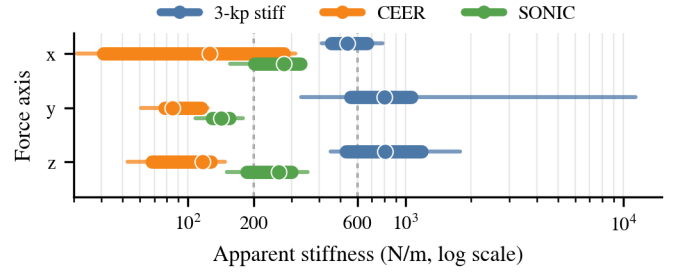


Fig. 3: Natural apparent stiffness of representative low-level policies. Positive and negative directions are combined for each axis. Thin lines show P5–P95, thick lines show the interquartile range, and dots show the median. Dotted lines indicate 200 and 600 N/m.

B. Low-Level Policy Characterization

Before evaluating the high-level controller, we characterize our stiff 3-kp low-level backbone against two representative alternatives: a compliant policy CEER [6] and a general motion-tracking policy SONIC [1]. We also vary the maximum training force of the 3-kp policy from 30 to 60 N. The results identify our 30-N 3-kp policy as the most suitable backbone, because it has low nominal tracking error with high apparent stiffness.

As shown in Table II, the stiff 3-kp policy provides the lowest nominal error and a median apparent stiffness near 600 N/m, making it a suitable low-level policy for high-level compliance training. Increasing the maximum training force from 30 to 60 N does not monotonically increase apparent stiffness and degrades nominal tracking, most clearly at 60 N. We therefore use the 30 N policy in the main experiments.

Figure 3 compares the natural Cartesian response of the three representative low-level policies, with measurements grouped by perturbation axis for clarity. Their apparent-stiffness distributions differ substantially across policies. CEER is naturally more compliant, while SONIC exhibits an intermediate response. The broad, uncommanded stiffness distributions motivate explicit high-level compliance control.

We select 600 N/m as the upper command bound because it is close to the overall median apparent stiffness of the low-level policy. The lower bound is constrained by reachable EE displacement: the curated AMASS motion corpus [27] has P5–P95 EE position widths above 0.4 m, while $K = 100$ N/m under the maximum evaluated force of 30 N requires a 0.30 m displacement. We therefore set the soft-axis residual limit to 0.35 m, providing a 0.05 m margin while remaining within the observed EE range. We use the 100–600 N/m command range, treating 100 N/m as its

TABLE III: Fixed-stiffness compliance evaluation at 200 N/m.

Policy	EE Comp. Err. (m) ↓	Nom. EE Err. (m) ↓	App. Stiffness (N/m)
Analytical MoE	0.023 ± 0.016	0.049 ± 0.045	219.6 ± 49.2
Fixed-stiffness HL	0.028 ± 0.014	0.050 ± 0.040	241.5 ± 55.7
Fixed-stiffness E2E	0.082 ± 0.034	0.100 ± 0.046	316.6 ± 512.0

workspace-limited lower bound.

C. End-Effector Compliance Evaluation

We evaluate hierarchy design, full-range directional compliance, and workspace consistency. The results show that the hierarchy reduces compliance-tracking error, the analytical MoE yields the lowest full-range errors, and errors remain similar across the tested EE poses.

1) *Fixed-Target Hierarchy Ablation*: We first evaluate whether separating compliance control from whole-body motion tracking improves accuracy at a single target stiffness. All methods in this experiment are trained or commanded to realize an isotropic stiffness of 200 N/m.

Analytical MoE is the full proposed method. **Fixed-stiffness HL** replaces the expert composition with a single high-level residual policy trained only for the 200 N/m target. **Fixed-stiffness E2E** removes the hierarchy and directly trains a single whole-body policy for the same target.

Table III shows that the analytical MoE and fixed-stiffness HL policy achieve EE compliance-tracking errors of 0.023 ± 0.016 m and 0.028 ± 0.014 m, respectively, compared with 0.082 ± 0.034 m for fixed-stiffness E2E. The two hierarchical methods also achieve apparent stiffness values closer to 200 N/m. The hierarchy therefore improves fixed-stiffness EE compliance, while analytical compliance-space composition preserves the accuracy of a dedicated fixed policy.

2) *Full-Range Directional EE Compliance*: We evaluate 16 stiffness configurations spanning the complete command range: three settings with one soft axis at 100 N/m and the other two at 600 N/m; three settings with one stiff axis at 600 N/m and the other two at 100 N/m; four isotropic settings at 100, 200, 400, and 600 N/m; and six permutations of 150, 300, and 500 N/m across the Cartesian axes. These settings test cross-axis expert composition, with the six non-anchor permutations additionally testing interpolation on all three axes. For each configuration, we follow the perturbation protocol in Sec. IV-A, resulting in 60 trials per configuration and 960 trials for each method in total.

a) *Baselines*: **Stiffness-conditioned HL** replaces the analytical MoE with one stiffness-conditioned high-level residual policy over the same stiff low-level policy. **Stiffness-conditioned E2E** trains a single stiffness-conditioned whole-body policy without the hierarchy. Both learned baselines are trained until convergence. The **oracle-force LL** baseline uses the ground-truth force to compute an EE command offset, $\Delta \mathbf{x}^* = \mathbf{K}^{-1} \mathbf{F}_{\text{ext}}$, which is added to the nominal EE command. It tests whether analytical force-to-offset conversion alone is sufficient when force-estimation error is removed.

b) *Metrics*: For trial i , apparent stiffness is $\hat{K}_i = \|\mathbf{F}_i\| / \|\mathbf{d}_i^\top \Delta \mathbf{p}_i\|$, where $\mathbf{d}_i$ is the unit applied-force direction

TABLE IV: Full-range directional EE compliance over 16 stiffness configurations (ideal trend slope and R^2 : 1).

Method	Accuracy			Trend	
	Compliance Matrix Err. ↓	EE Comp. Err. (m) ↓	Stiffness MAPE ↓	Slope → 1	R^2 ↑
Analytical MoE	0.278	0.0349	0.249	0.764	0.779
Stiff.-cond. HL	0.677	0.0566	1.325	0.136	0.040
Stiff.-cond. E2E	1.064	0.0884	5.208	3.430	0.003
Oracle-force LL	0.304	0.0459	0.256	0.485	0.792

and $\Delta \mathbf{p}_i$ is the EE displacement from the unperturbed baseline. Stiffness MAPE is $N^{-1} \sum_i |\hat{K}_i - K_i^{\text{cmd}}| / K_i^{\text{cmd}}$, where $N = 960$ is the number of trials per method and K_i^{cmd} is the commanded stiffness along the applied-force direction. To capture directional coupling, we fit $\Delta \mathbf{x} = \mathbf{b} + \hat{\mathbf{C}} \mathbf{F}$ by least squares and report the normalized compliance-matrix error $\|\hat{\mathbf{C}} - \mathbf{C}^*\|_F / \|\mathbf{C}^*\|_F$, where $\mathbf{C}^* = \text{diag}(1/K_x, 1/K_y, 1/K_z)$; this error is averaged over both hands and all stiffness configurations. Finally, a global ordinary-least-squares fit with an intercept, $\hat{K}_{\text{measured}} = \alpha K_{\text{commanded}} + \beta$, gives the trend slope α and R^2 , both ideally 1.

c) *Results*: Table IV shows that the analytical MoE achieves the lowest compliance-matrix error (0.278), EE compliance-tracking error (0.0349 m), and stiffness MAPE (0.249), with the slope closest to 1. As shown in Fig. 4, the stiffness-conditioned HL policy maps the full command range to much narrower, axis-dependent response ranges. Its low global slope (0.136) quantifies this range compression, while its low R^2 (0.040) indicates poor command-response consistency. The stiffness-conditioned E2E policy further degrades accuracy and has nearly zero R^2 .

This range compression is consistent with multi-task interference: a single stiffness-conditioned policy can optimize a compromise response rather than preserve distinct axis-specific behaviors across the command range. End-to-end training further couples compliance learning with whole-body motion tracking. The analytical MoE separates these objectives and composes specialized EE compliance behaviors, reducing both sources of interference.

The oracle-force LL baseline produces a consistent trend ($R^2 = 0.792$), but its slope of 0.485 and EE compliance-tracking error of 0.0459 m remain worse than those of the analytical MoE. Directly applying $\mathbf{K}^{-1} \mathbf{F}_{\text{ext}}$ assumes that the low-level policy is an ideal Cartesian position controller. Its finite, unregulated apparent stiffness and coupled whole-body dynamics violate this assumption, so the requested displacement is not realized exactly. Notably, our method outperforms the oracle-force baseline despite using estimated rather than ground-truth forces, providing evidence that the learned EE compliance experts *compensate* for the low-level closed-loop dynamics, not merely infer the external force.

3) *Workspace Consistency*: We evaluate workspace consistency at seven paired EE poses: the center pose and 0.05 m offsets along each positive and negative Cartesian axis. At each pose, we apply the full 16-configuration protocol, totaling 6,720 trials. For each hand, pose, and Cartesian component, we average the absolute difference between the

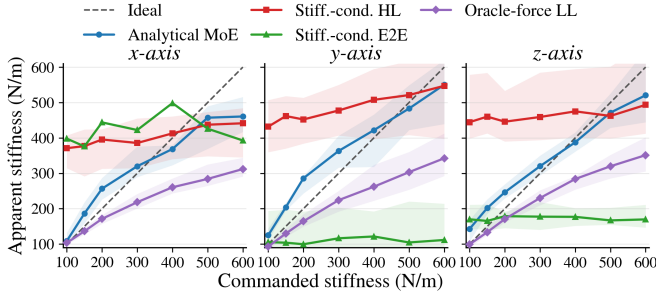


Fig. 4: Commanded versus apparent stiffness over the 16-configuration 100–600 N/m evaluation, shown separately for the x , y , and z axes. Curves show the median response, shaded regions show the interquartile range, and the dashed line denotes ideal tracking. For readability, the unusually broad stiffness-conditioned E2E IQR band is omitted only in the x panel.

TABLE V: Root-compliance evaluation using the 3-kp root-residual action space without yaw control.

Mode	Root drift (m) ↓	Velocity error (m/s) ↓	Apparent B median / target
3-kp LL only	0.3627	0.0955	–
Resistance	0.3103	0.0987	–
$B = 60$	1.7806	0.1143	57.0 / 60
$B = 200$	0.7163	0.1031	201.4 / 200

actual and target EE displacements over 480 trials. Across the tested poses, the mean x -, y -, and z -component errors are 24.8, 22.1, and 25.8 mm, respectively. The largest pose-wise mean component error is 43.7 mm at the raised-hand $z+$ pose, which may reflect reduced joint-limit margin or kinematic redundancy when the arms are elevated. The two hands show similar qualitative trends but retain quantitative asymmetries: the left x response is generally less accurate, whereas the right z response is generally less accurate.

D. Root Compliance Evaluation

We next evaluate whether the root compliance policies produce distinct resistance and damping responses. Forces of $\{5, 10, 20, 30\}$ N are applied along four horizontal directions at the torso, shoulders, wrists, and hands, giving 112 cases per policy. We study three modes. **Resistance** minimizes root displacement and uses zero target velocity. The two damping modes target an along-force velocity change $\Delta \mathbf{v}^* = \mathbf{F}_{xy}/B$ with $B = 60$ or 200 N s/m. Planar root drift measures position retention, while $\|\Delta \mathbf{v} - \Delta \mathbf{v}^*\|$ measures the root velocity-response error.

Table V shows that the three root compliance modes yield distinct responses. The resistance mode moves least, $B = 60$ follows the pull most, and $B = 200$ gives an intermediate response. The median apparent damping closely matches both commanded values.

We further test whether additional locomotion-relevant control authority improves the three root modes. As shown in Table VI, foot and yaw residuals substantially improve the resistance mode, where active turning and stepping reject disturbances. Their improvements in the damping modes are smaller and configuration-dependent. Thus, additional control authority helps stabilization but does not fundamentally

TABLE VI: Root-policy action-space ablation. Resistance is evaluated by planar root drift (m), and damping by velocity-response error (m/s). All variants include root-command residuals; 5-kp adds foot-position residuals, and +yaw adds a root-yaw residual.

Mode	3-kp	3-kp+yaw	5-kp	5-kp+yaw
Resistance	0.3103	0.2032	0.4964	0.1574
$B = 60$	0.1143	0.1217	0.1094	0.1073
$B = 200$	0.1031	0.0678	0.0904	0.0700

resolve contact- and gait-dependent damping control.

E. Real-World Evaluation of EE and Root Compliance

We directly transfer all policies from simulation to real and evaluate tunable Cartesian stiffness, directional EE compliance, and combined EE–root compliance in real-world tasks, including online adjustment of EE stiffness. Because manual force and displacement measurements yield only approximate stiffness estimates, we focus primarily on task-level performance. Fig. 1 provides an overview, while Fig. 5 summarizes the trials. Complete experiments appear in the supplementary video and on the project website.

1) *Tunable EE Compliance and Online Stiffness Adjustment*: We estimate the real-world end-effector stiffness by pulling the EE with a force gauge. The ruler reading changes from 1 to 3 in, giving a net displacement of 2 in. Under $K_y = 100, 200$, and 600 N/m, the measured forces are 10.9, 17.2, and 28.0 N, respectively, corresponding to apparent stiffness values of 214.6, 338.6, and 551 N/m (Fig. 5(m–o)). The remaining discrepancy is reasonable given the uncertainty in the manual force and displacement measurements. In yoga-ball grasping, soft and stiff y -axis settings produce visibly different ball indentations (Fig. 5(p,q)). In payload dragging, increasing K_z reduces the vertical EE displacement under the same 3 kg payload (Fig. 1(b)). The EE stiffness is adjusted online without restarting the low-level policy.

2) *Directional EE Compliance*: Writing requires low surface-normal stiffness K_z and high tangential stiffness (K_x, K_y) . When writing “8” (Fig. 5(a–c)), the isotropically soft policy often loses contact, whereas the directional $K_z = 100$ N/m and isotropically stiff settings complete the trajectory. The isotropically stiff setting, however, breaks the pen holder in another trial (Fig. 1(c)). We then draw a straight line while applying a disturbance perpendicular to the desired line (Fig. 5(d–f)). Increasing stiffness along the disturbance direction visibly reduces the resulting protrusion in the drawn trajectory. On a surface tilted by 10° , panels (g) and (h) compare whether the pen tilts under different directional settings, while the line in panel (j) is less straight than that in panel (k). Isotropically high stiffness again breaks the holder. Together, these results demonstrate the need for directional EE compliance: low surface-normal stiffness preserves safe contact, while high tangential stiffness improves trajectory accuracy.

3) *Combined Directional EE and Root Compliance*: We verified the usefulness of combined directional EE and root compliance via collaborative box carrying, a human-robot co-manipulation task. The EE is compliant along the contact

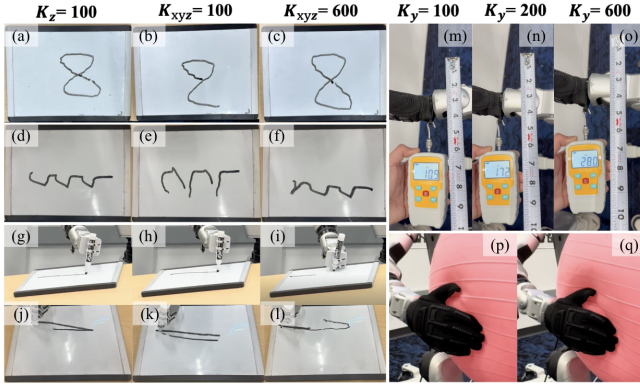


Fig. 5: Directional EE compliance tasks: writing “8” (a–c), straight-line writing under disturbance (d–f), and writing on a tilted surface (g–i). Tunable EE-stiffness demonstrations: force-gauge pulling (m–o) and yoga-ball grasping (p–q).

y -axis and stiff along x and z , while the root uses the $B = 60$ damping mode to follow the human interaction forces. With a 3 kg payload, this combination completes the task, whereas isotropically low EE stiffness cannot maintain support and the box slips (Fig. 1(a)). This task requires directional EE compliance to accommodate interaction while supporting the payload and root damping to follow the human, highlighting the need for combined EE–root compliance.

V. CONCLUSION

We presented a structured formulation of humanoid compliance that combines directional and tunable EE compliance with root damping and resistance. A two-level hierarchical RL controller with an analytical compliance control MoE realizes these behaviors on top of a shared stiff low-level whole-body tracking policy, supporting online adjustment of EE stiffness and root mode switching. Simulation evaluations and real-world loco-manipulation experiments show that different compliance configurations enable safer contact, more accurate manipulation, payload adaptation, and collaborative physical interaction.

The current implementation has several limitations. First, the achievable compliance behavior is limited by the capability of the stiff low-level whole-body tracking policy. Since the high-level controller only modulates its EE–root commands, weaknesses of the underlying tracker directly carry over to the compliance controller. In particular, insufficient x -direction stiffness limits the achievable EE stiffness range, while limited locomotion capability constrains root-level interaction performance. Second, although the framework supports three root compliance modes, physical interaction is not yet sufficiently smooth. The robot often takes heavy steps and lacks deliberate foot-placement adaptation under interaction. Future work will improve root compliance by explicitly incorporating interaction-dependent foot-placement targets and gait regularization that penalizes large step impacts and abrupt contact transitions.

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